

```
$ retag run --taxonomy consolidated-2026 --workers 12
```

```
scanning content items ..... 4,182 found  
loading taxonomy v41 ..... 312 terms  
12 tenant workers started  
proposing ..... 4,182 / 4,182  
applying ..... 4,182 updated
```

```
TENANTS: 12  
UPDATED: 4,182  
ERRORS: 0  
DURATION: 00:41:16    22:40 → 23:21
```

```
done.
```



When Workflows Meet Meet Agents

Agents supply judgment. Workflows retain authority.

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demo repo



Two things I'm not claiming

1 ~~I'm here to teach you Orchard~~

I'm mapping operational agent failures onto primitives you already know. You'll know. You'll catch things. Say them out loud.

2 ~~Twenty cases is coverage~~

It's a canary. Most teams have no canary at all.

Three weeks later

TICKET #8841 priority: low
tenant: north-shore-council

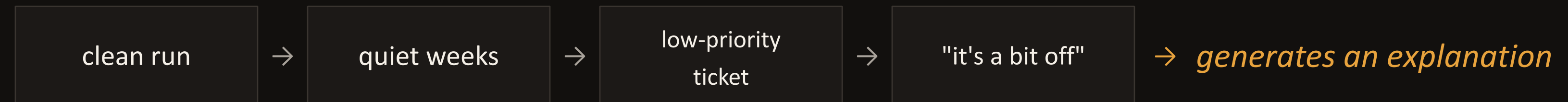
"Why is our Emergency Services page filed under
Community Events? It's been like that a while."

Exceptions fail loudly. Judgments fail plausibly.

EXCEPTION

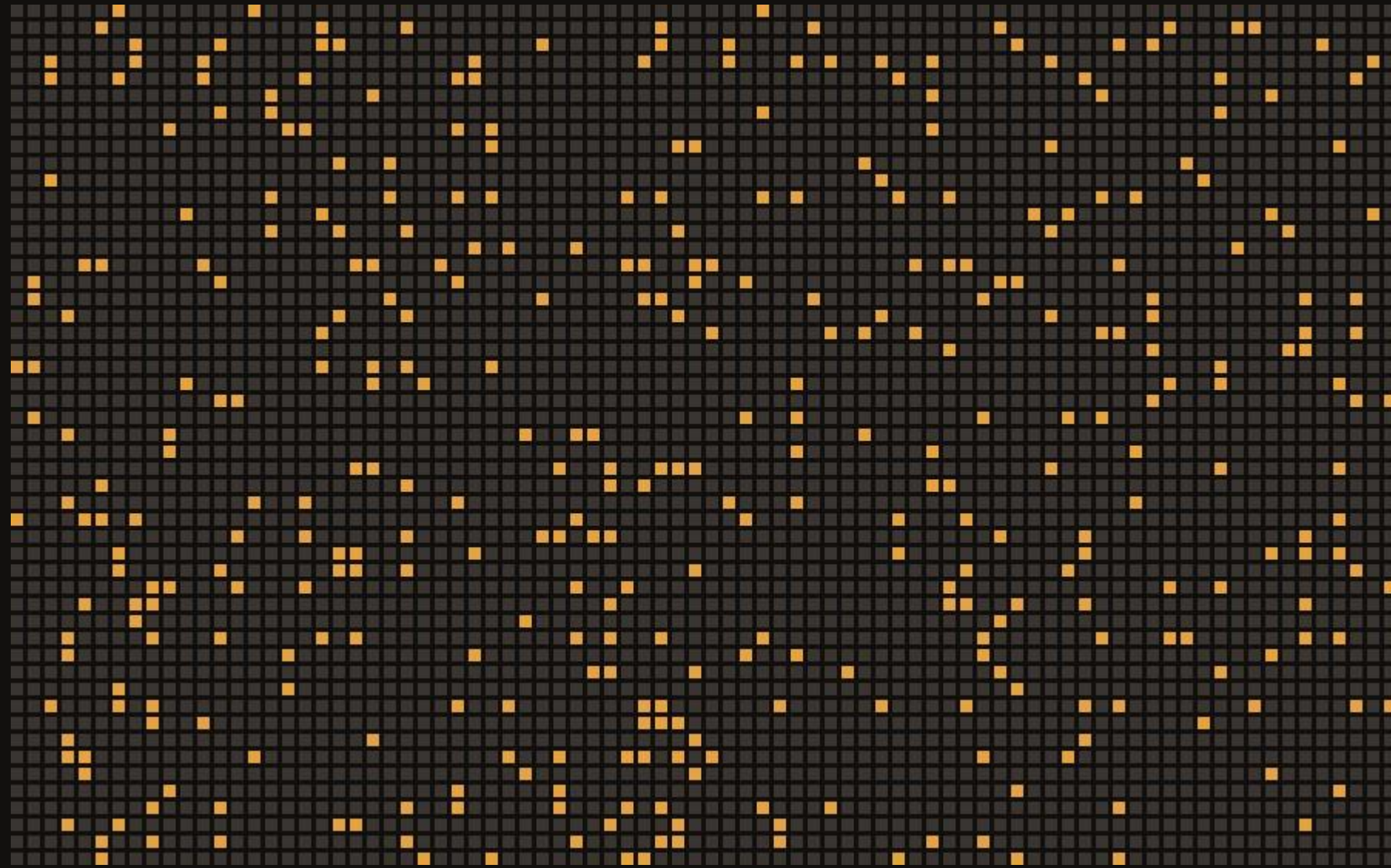


JUDGMENT



The support ticket closes. The system defect survives.

Rollback is not the problem



every square passed validation.

Versioning makes recovery possible. Provenance makes recovery selective.

23:10

22:40 12 workers start · taxonomy v41

- workers mid-batch stay on v41
- later batches start on v42

23:21 complete · 0 errors

23:10 editor publishes taxonomy v42

v41

emergency-preparedness
→ Community Events

term#0e5c

v42

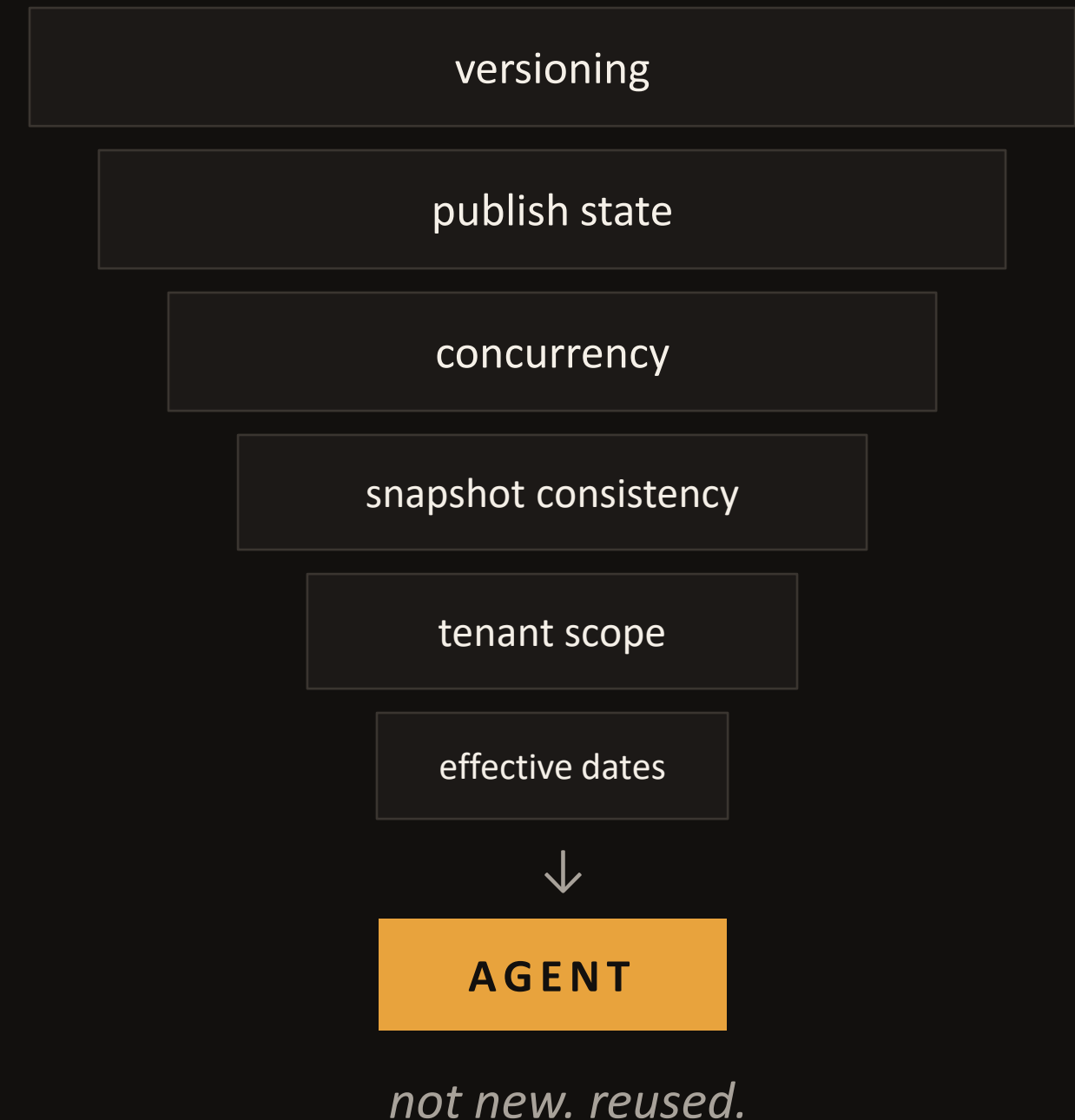
emergency-preparedness
→ Public Safety

term#0e5c

guidance different · identifier identical

3,082 on v41 · 1,100 on v42 · writes interleaved

Content-lifecycle failures in a costume



*Your agent is the least careful consumer your content
content store has ever had.*

The model recommends. Code verifies. verifies.

KEEP IT CODE

required fields

permitted types

term exists

term set version is current

agent release version is pinned

tenant scope

publish permission

lot size

WORTH A MODEL

what is this page about

reconciling sources that disagree

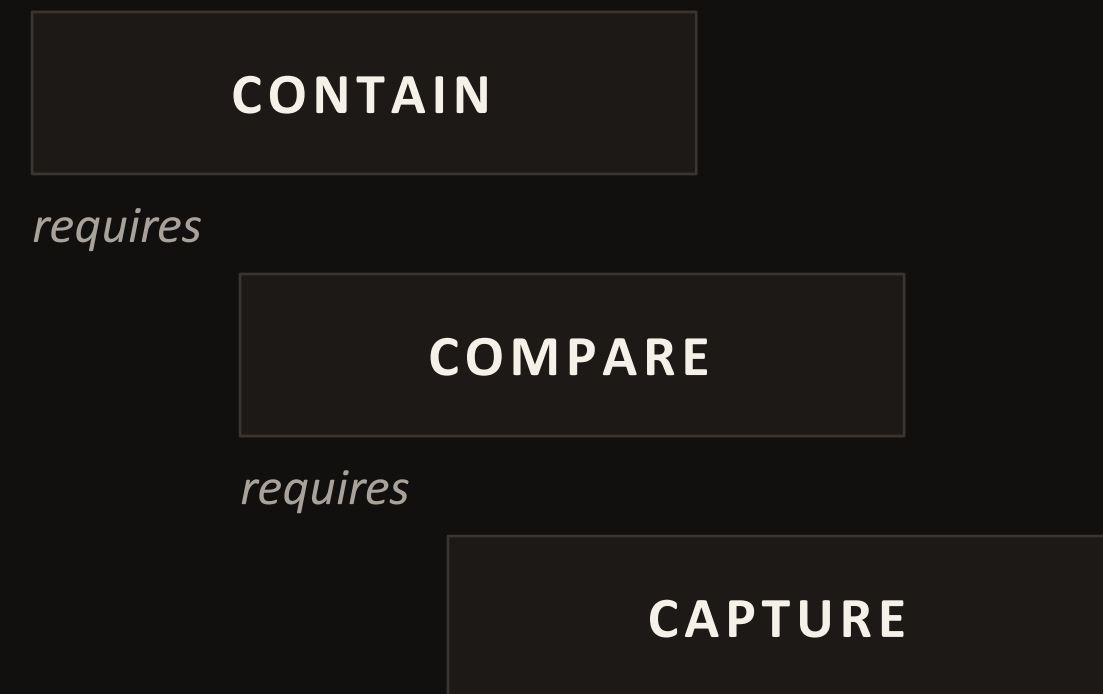
summarising without losing the point

register in translation

genuine ambiguity

What you want is the reverse of what you build

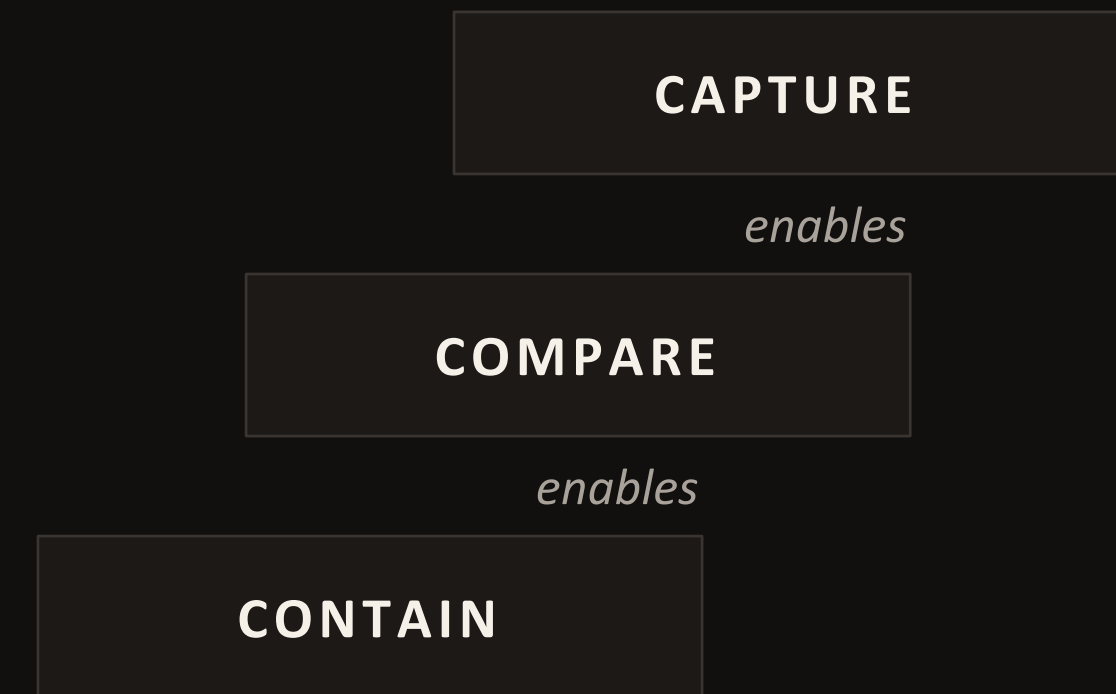
what you want



OPPOSITE



what you build



A kill switch you never trip is theater.

CAPTURE

COMPARE

CONTAIN

Check my work

three of these are probably wrong.

what an agent needs around it

Hold a change without committing it

Undo a change you regret

Survive a restart mid-process

Pause for a human, resume on their answer

Branch on a result

Limit what data is reachable

Decide who may act

The point of no return

A record of what changed and by whom

WHAT YOU ALREADY CALL IT

A draft version

Version history

A workflow instance

A signal

Outcomes and transitions

Tenants

Roles and permissions

Publish

Audit Trail

The agent owns one box

Load tenant context



Pin taxonomy version + agent release version



Deterministic validation



Evaluation



Approval / policy outcome



Commit the exact proposed version

Orchard owns business state, progression, approval and commit.

The agent runtime owns what happens inside the box.

The contract you can't write

CONTENT TYPE

- ✓ required fields
- ✓ field types
- ✓ term exists
- ✓ tenant scope

fires on save.

THE JUDGMENT

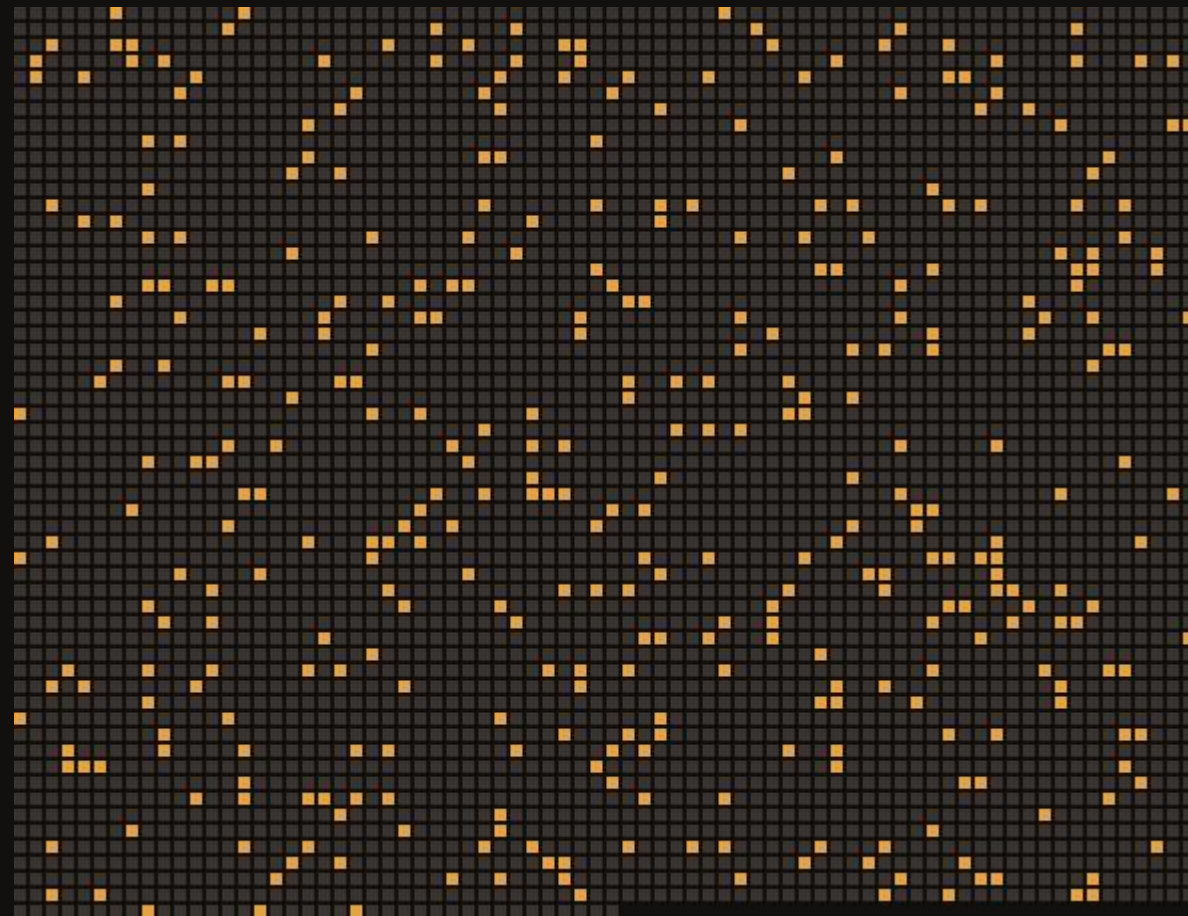


saves clean, every time.

Evaluations are the contract for the half you can't validate.

Blast radius is a design parameter

--workers 12, unbounded



*first check:
3 weeks.*

--lot tenant --canary 40



✓ first check: minutes.

Nobody chose 4,182. It was just how many there were.

A trace is just another table

SPAN

trace_id

span_id

parent_id

name

duration_ms

status

attributes



*same rows.
familiar joins.*

```
select name,  
       count(*) as calls,  
       avg(duration_ms) as avg_ms  
from   spans  
where  job_id = 'retag-2026'  
group by name  
order by avg_ms desc;
```

OpenTelemetry → somewhere queryable → the SQL you already write.

Two mutable inputs. Pin both.

TAXONOMY SNAPSHOT

taxonomy.version: 41

published by an editor

changes without a deployment

=

AGENT RELEASE 17

instructions hash

model + settings

tool schema hash

retrieval config

policy version

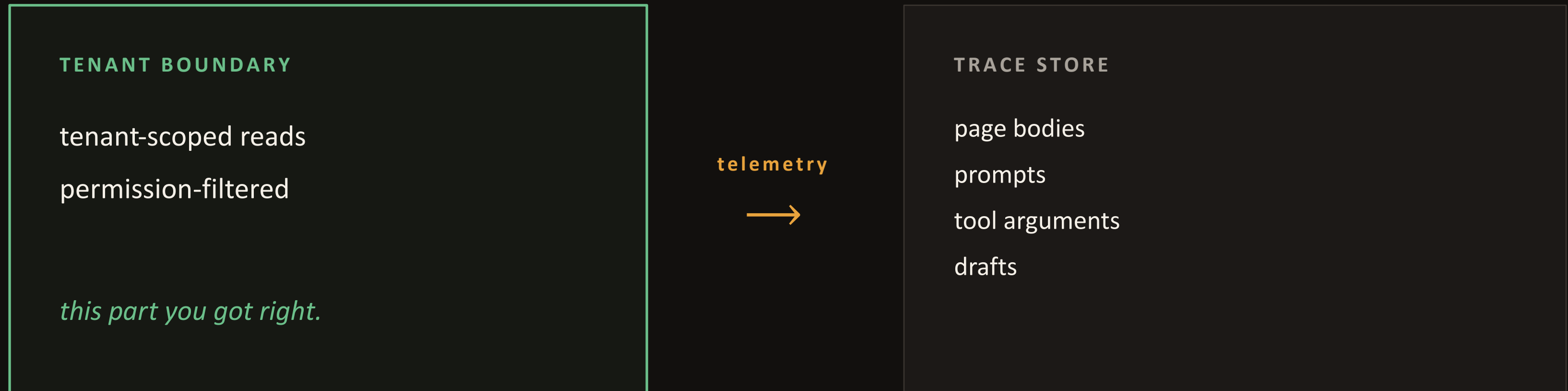
eval suite

changes without a deployment

both are inputs. Both must be pinned, captured, and asserted before commit.

tenant · item · version · locale · job id · chosen terms ·
considered · taxonomy.version · agent.release ·
eval results · policy outcome · committed version

Your trace store is a content repository



who can read it · how long is it kept · what is redacted before it leaves the process

\$ retag trace show item-31882

```
span propose.classify
├─ considered  Community Events
│   Public Safety
│   Civil Contingency
│   Local Services
├─ chose      Community Events
├─ rationale  "page describes local
│   emergency contacts"
└─ taxonomy.version: 41
```

\$ retag traces group-by taxonomy.version

version	decisions	status
-----	-----	-----
42	1,100	current
41	3,082	SUPERSEDED ◀
-----	-----	-----
	4,182	

→ 3,082 item ids written to replay-v41.txt

\$ retag replay --from replay-v41.txt --taxonomy 42

```
replaying 3,082 decisions under v42 ..... done
proposals identical ..... 2,702
proposals differ ..... 380 ◀
```

→ 380 item ids written to compensate.txt

Accuracy is the wrong number

DECISIONS NOT ON CURRENT VERSION	3,082	(73.7%)
KNOWN-ANSWER PASS RATE	66.7%	falling
HELD BY POLICY	1	active
TECHNICAL ERRORS	0	flat

*one hold. zero errors.
working as designed.*

Accuracy is a number you compute after somebody establishes the
establishes the answer.

Not all evals assert on the answer

TYPE	ASSERTS ON	NEEDS EXECUTION EVIDENCE?
Outcome	The final classification	No
Process	What it considered, what it chose	the trajectory is the dataset
Efficiency	Tokens, cost per item, duration	the trajectory is the dataset
Source-layer	Invariants about the taxonomy itself	No
Cross-signal	Two signals that can’ t both be true	Sometimes
Judge	Tone, completeness, answered the brief	Sometimes

You cannot evaluate what you did not capture.

Twenty cases you already have

A HUMAN ALREADY DID THESE

the two hundred items re-tagged by hand before anyone wrote the agent

SOMEBODY COMPLAINED

ticket 8841 and the items around it

EMBARRASSINGLY OBVIOUS

Bin Collection Schedule is Waste Services, or something is something is very wrong

YOU ARGUED ABOUT THESE

the genuinely ambiguous ones your editors debated. Record
Record the verdict.

Not from imagination. An afternoon, not a quarter.

Exact-match contracts. Rubric-score prose.

✓ EXACT MATCH

term_ids

taxonomy.version

agent.release

tenant

deprecated: false

✓ RUBRIC SCORE

groundedness

completeness

tone

answered the brief

REGENERATE WHEN

the business rule changed

you changed the model on purpose

a case is obsolete

NEVER REGENERATE BECAUSE

the suite went red and you want it green

that's not a golden set. That's a screenshot of the bug.

AGENT RELEASE 17 → 18 instructions:

+ Prefer broader terms. Editors find deep
+ hierarchies confusing.

\$ retag run --lot tenant --canary 40 --release 18

tenant 1 canary 40 · 20/20 · COMMITTED 428 items
tenant 2 canary 40 · 20/20 · COMMITTED 356 items
tenant 3 canary 40 · 20/20 · COMMITTED 511 items
tenant 4 canary 40 ·

OBVIOUS	[REDACTED]	6/6	PASS
HUMAN-VERIFIED	[REDACTED]	6/6	PASS
ARGUED	[REDACTED]	2/8	FAIL ◀

14/20 — every failure chose the parent term

➤ CANARY HELD. Proposals remain drafts.
0 items committed for this tenant.

ERRORS: 0 HELD BY POLICY: 1

Gates everywhere means gates nowhere

✓ GATE IT

irreversible

reaches a customer

crosses a tenant

changes permissions

a lot about to commit

DON'T

~~reads~~

~~drafts~~

~~proposals~~

~~a single reversible item~~

4,182 approvals is not a control. It's a resignation letter.

Twelve approvals. One per tenant.

CAPTURE

COMPARE

CONTAIN

Three layers. The middle one is yours.

1 PROMPT

"Always stop at tenant boundaries."

a request. it may comply.

2 WORKFLOW

an outcome, a signal, a permission check

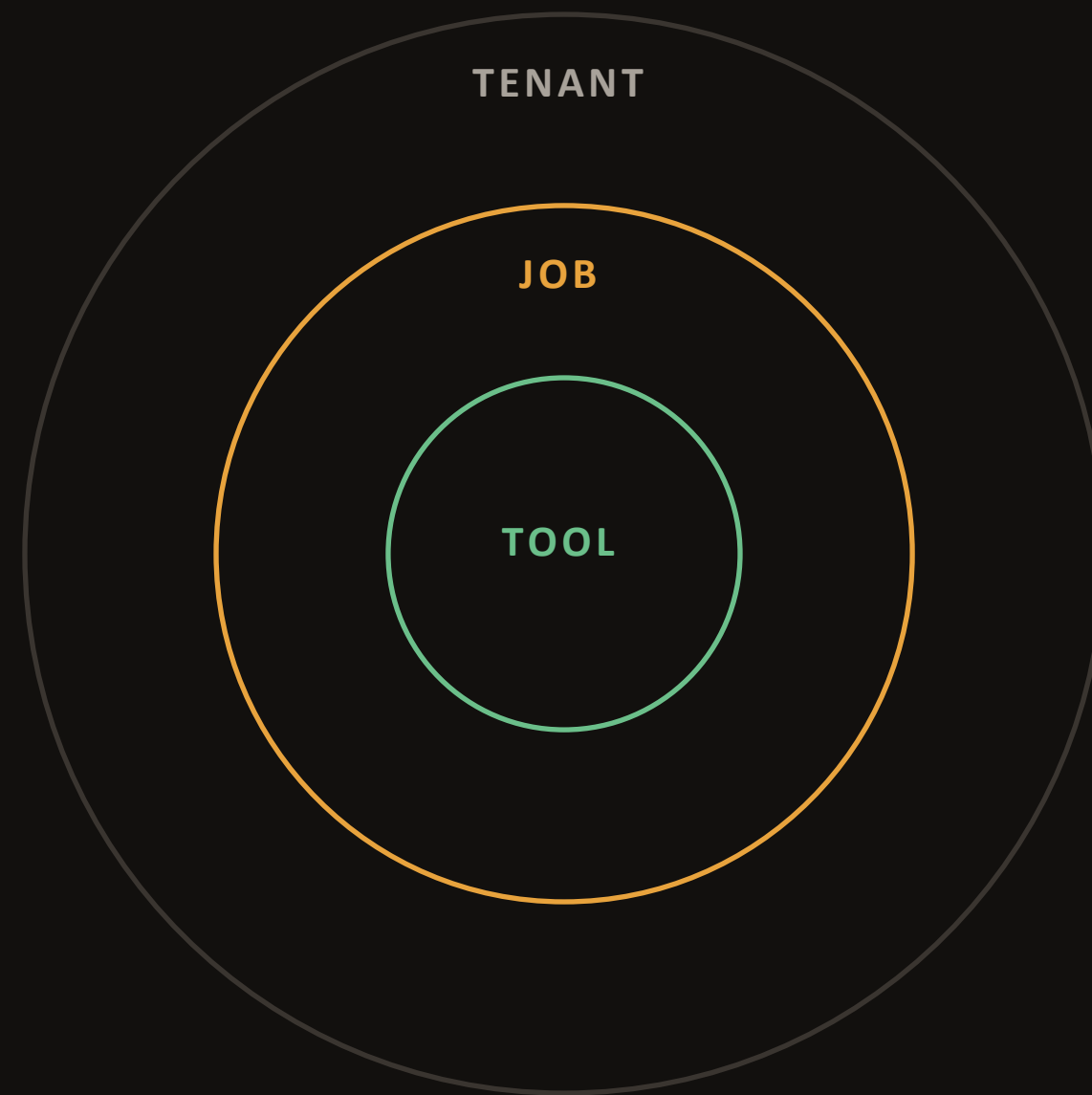
*business authority.
this is where it belongs.*

3 RUNTIME

```
var commit = AIFunctionFactory.Create(CommitLot);  
var gated = new ApprovalRequiredAIFunction(commit);
```

*// the run complet
execution-layer backstop.*

Three levels of stop



TOOL

disable one capability. The job keeps running.

You'll use this most. Nobody builds it.

JOB

this run stops. Everything else keeps running.

TENANT

all automation stops for one customer.

Roughly never. Glad it exists that day.

In the demo, all three are a flag file and an if-statement.

\$ retag status --job retag-2026

tenant 4 · lot complete · 389 items proposed
canary 14/20 ARGUED 2/8
commit GATED — awaiting approval
decision pending · editorial lead

➤ 0 items committed. Proposals remain drafts.

ERRORS:0 | HELD BY POLICY:1



different columns, on purpose.

\$ retag stop --job retag-2026

writing stop flag ok
pending lots cancelled 8
in-flight lot drained 1

job retag-2026 STOPPED
job seo-audit running
job link-check running

Same job. Twelve lots.

pin taxonomy version + agent release



tenant lot 1 · canary 40



20 known-answer cases ◀



hold before first live commit ◀



remainder of tenant, bounded chunks
versions asserted before each commit



repeat × 12

before — 41 min · 0 errors · 3 weeks to detection · 1 week to diagnose · 9 tenants affected

after — 48 min · 0 errors · minutes to detection · 1 query to diagnose · 1 tenant maximum

Locate yourself. Honestly.

- 4 Invariants catch drift before the agent consumes it
- 3 Failures trigger controls without a human deciding
- 2 Execution evidence lands somewhere queryable
- 1 Twenty cases with the expected decision

Built bottom-up. Climbed top-down. Deliberate.



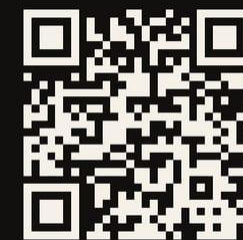
Write down twenty cases.
Write down the expected decision.

that's rung one. an afternoon.

That's not coverage. It's a canary. Most teams have no canary at all.



demo repo
console app, cases, invariants



LinkedIn
tell me what broke

Static fallback panels

span propose.classify
└─ chose Community Events
└─ rationale "local emergency
| contacts"
└─ taxonomy.version: 41

1 THE VERSION ATTRIBUTE

2 THE GROUP-BY

taxonomy.version: 41

version	decisions	status
42	1,100	current
41	3,082	SUPERSEDED ◀

3 THE HELD CANARY

OBVIOUS [] 6/6 PASS HUMAN-VERIFIED [] 6/6 PASS ARGUED [] HELD 2/8 POLICEX 11 ◀

4 THE TWO-COLUMN FOOTER

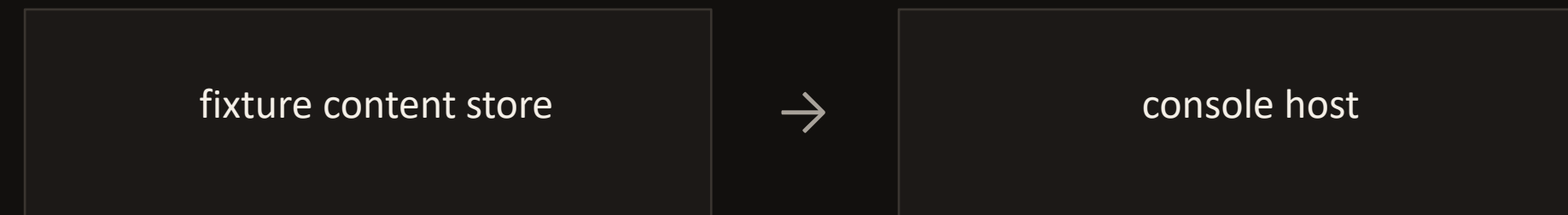
Different columns, on purpose. A rising held rate is information — not information — not a defect, and not in your error budget.

14/20 — every failure chose the parent term

➤ CANARY HELD. Proposals remain drafts.
Narrate identically to the recording. Build these regardless of demo progress.

Reference architecture

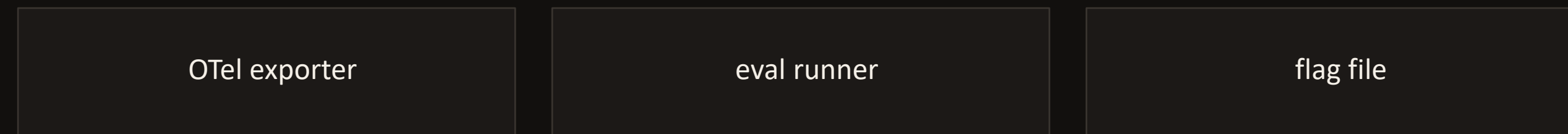
From stage this reads as a vendor pitch. Keep it off the stage.



THREE TOOLS



EVIDENCE AND CONTROL



Framework doesn't matter. Instrumentation does.

The proposal record

```
public sealed record TagProposal(  
    string ContentItemId,  
    string TenantId,  
    IReadOnlyList<string> TermIds,  
    int TaxonomyVersion,  
    string AgentRelease,  
    IReadOnlyList<string> Considered,  
    string Rationale,  
    string TraceId);
```

Don't return a string and make the caller infer what happened.

Agent Control Specification

public preview · 0.3.1-beta

Policy layer of the Agent Governance Toolkit, vendored as its policy engine

Stateless, deterministic, fail-closed policy decision runtime

Rust core with bindings for .NET, Python, Node and Rust

The host is the enforcement point — ACS returns a verdict, the host applies it

Verdicts: allow, warn, deny, escalate, transform

Policies are Rego bundles, not simple declarative YAML

Do not cite a checkpoint count. Say “across the agent loop” and leave the number alone.